Curriculum Vitae

Jinho Byun

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RESEARCH EXPERIENCE

Mar. 2025 – Present	Research Professor, Dept. of Energy Engineering, KENTECH
Mar. 2023 – Mar. 2025	Postdoctoral Researcher, Dept. of Energy Engineering, KENTECH
Aug. 2022 – Jan. 2023	R&D Engineer, Wonik IPS
Mar. 2022 – Jul. 2022	Postdoctoral Researcher, Dept. of Physics, Pusan National University (Advisor: Prof. Jaekwang Lee)

EDUCATION

Sep. 2017 – Feb. 2022	Ph.D., Physics, Pusan National University. Advisor: Prof. Jaekwang Lee. Thesis: "Emergent Phenomena at Polar Oxides and Their Applications."
2012 – 2017	B.A., Physics, Pusan National University

HONORS AND AWARDS

2021	Outstanding Poster Award, Korean Physical Society (KPS) — "Interface engineered TER optimization in HfO2-based FTJ"
2021	Excellent Academic Award, Pusan National University
2020	Humantech Paper Award, Samsung Electronics — "Anisotropic Evaporation of ZnO Observed by In-situ Cs-Corrected High Resolution TEM"
2019	Outstanding Poster Award, Korea Institute for Advanced Study (KIAS) — "First-principles study of the HfO2-based ferroelectric tunnel junction"
2017	Outstanding Presentation Award, Korean Physical Society (KPS) — "Design of high-coercivity Fe16-xAlxN2 alloy"
2017 – 2020	Combined-course Scholarship (full tuition), Pusan National University
2012 – 2017	National Excellence Scholarship (Natural Sciences and Engineering), Korean Government (full tuition)

RESEARCH INTEREST

Methods: deep learning for microscopy, ab initio multislice image simulation, ab initio / machine-learning-force-field molecular dynamics, phonon properties, DFT

Materials systems: oxide surfaces and interfaces, perovskites, 2D materials

PUBLICATIONS

¶ equal/co-first contribution; * corresponding author.

Accepted / in press

1. Y. Xing¶, **J. Byun¶**, K. T. Kang, J. Hwang, J. Lee, W. S. Choi, S. H. Oh*. "Atomic-Scale Visualization of Vacancy-Ordered Intermediates and Pathway Selection in the Perovskite–Infinite-Layer Transformation." *Journal of the American Chemical Society* (2026). **Accepted; in production.** doi:10.1021/jacs.6c08044

Under review

2. K. Lee¶, Y. Xing¶, **J. Byun¶**, M. Jeong¶, J. Seo, J. Lee, J. Bang, E. Lee*, S. H. Oh*. “Visualizing Oxygen Vacancy-Induced Dielectric Dead Layer at Metal-Oxide Interfaces by In-situ Multimodal Electron Microscopy.” *Science Advances* (under review, major revision).
3. J. Jeong¶, **J. Byun¶**, X. Xu, A. Gruverman, J. Lee, S. H. Oh*. “Direct Visualization of Interfacial Defect Effects on Polarization Switching in BaTiO3 Tunnel Junctions.” *Science Advances* (accepted, pending final technical revision).
4. Hojin Lee, Joonbong Lee, **J. Byun**, Sang-Hyeok Yang, Ayoung Cho, Sangwoo Lee, Dae Haa Ryu, Hyunbin Chung, Hee Seo Yun, Moon Seop Choi, Yesul Choi, Sungkyun Park, Min-Hyung Jung, Dongwon Lee, Yoon-Uk Heo, Hu Young Jeong, Y.-M. Kim*, J. Lee*, T. Choi*. “Interface-Engineered Hybrid Ferroelectric Tunnel Junctions Exhibiting Intrinsic Nonlinearity and Self-Rectification via Antiphase-Like Interlayer Formation.” *Advanced Functional Materials*, manuscript 7502704 (under review, second round).
5. Y. Xing¶, **J. Byun¶**, J. Seo, Z. Wang, Y. Jin, S. Kim, J. Lee, S. H. Oh*. “Geometry-driven surface electron dichotomy via ordered subsurface defects.” *Nature Physics* (under review).

Peer-reviewed journal articles (published)

6. Y. Xing, I. Kim, K. T. Kang, J. Byun, W. S. Choi, J. Lee, S. H. Oh*. “Monitoring the formation of infinite-layer transition metal oxides through in situ atomic-resolution electron microscopy.” *Nature Chemistry* **17**, 66–73 (2025). doi:10.1038/s41557-024-01617-7 [Author Correction: doi:10.1038/s41557-025-01816-w]
7. J. Seo, H. Lee, K. Eom, J. Byun, T. Min, J. Lee, K. Lee, C.-B. Eom, S. H. Oh. “Field-induced modulation of two-dimensional electron gas at LaAlO3/SrTiO3 interface by polar distortion of LaAlO3.” *Nature Communications* **15**, 5268 (2024). doi:10.1038/s41467-024-48946-2
8. J. Hwang, J. Byun, H. Kim, J. Lee. “High-mobility two-dimensional electron gas at the PbZr0.5Ti0.5O3/BaSnO3 heterostructure.” *J. Korean Phys. Soc.* **84**(1), 67–72 (2024). doi:10.1007/s40042-023-00978-5
9. Z. Wang¶, **J. Byun¶**, Z. Wang, Y. Xing, J. Seo, J. Lee, S. H. Oh*. “Direct Observation of Atomic Step-Assisted Stabilization of Polar Surfaces.” *Advanced Materials* **35**(40), 2303051 (2023). doi:10.1002/adma.202303051
10. M. H. Sheen, Y. H. Ko, D. U. Kim, N. R. Ahn, C. H. Lee, D. H. Kim, R. Kim, I. P. Kim, et al., J. Byun. “Light emitting element, method for fabricating the same and display device.” US Patent App. 18/093,959 (2023). [Patent — no DOI]
11. Z. Wang¶, **J. Byun¶**, S. Lee, J. Seo, B. Park, J. Kim, H. Y. Jeong, J. Bang, J. Lee, S. H. Oh*. “Vacancy driven surface disorder catalyzes anisotropic evaporation of polar ZnO (0001) surface.” *Nature Communications* **13**, 5616 (2022). doi:10.1038/s41467-022-33353-2
12. Y. Ha, J. Byun, J. Lee, J. Son, Y. Kim, S. Lee. “Infrared Transparent and Electromagnetic Shielding Correlated Metals via Lattice-Orbital-Charge Coupling.” *Nano Letters* **22**, 6573–6579 (2022). doi:10.1021/acs.nanolett.2c01487
13. M. Sheen, Y. Ko, D. Kim, J. Kim, J. Byun, K. Y. Yeon, D. Kim, J. Jung, J. Choi, R. Kim, J. Yoo, I. Kim, C. Joo, N. Hong, J. Lee, S. H. Jeon, S. H. Oh, J. Lee, N. Ahn*, C. Lee*. “Highly efficient blue InGaN nanoscale light-emitting diodes.” *Nature* **608**, 56–61 (2022). doi:10.1038/s41586-022-04933-5
14. K. Kang, J. Byun, M. Jeen, G. Jo, Y. K. Baek, J. Lee, H. Jeen. “Effect of sintering time on electronic properties of strontium hexaferrite.” *Ceramics International* **48**(9), 12476–12482 (2022). doi:10.1016/j.ceramint.2022.01.113
15. K. Lee, J. Byun, K. Park, S. Kang, M.-S. Song, J. Park, J. Lee, S. Chae*. “Giant Tunneling Electroresistance in Epitaxial Ferroelectric Ultrathin Films Directly Integrated on Si.” *Applied Materials Today* **26**, 101308 (2022). doi:10.1016/j.apmt.2021.101308
16. J. Lee, M. S. Song, W.-S. Jang, J. Byun, H. Lee, M. H. Park, J. Lee, Y.-M. Kim*, S. C. Chae*, T. Choi*. “Modulating the ferroelectricity of hafnium zirconium oxide ultrathin films via interface engineering to control

the oxygen vacancy distribution." *Advanced Materials Interfaces* **9**(7), 2101647 (2022). doi:10.1002/admi.202101647

17. Y. Ha, J. Byun, J. Lee, S. Lee. "Design Principles for the Enhanced Transparency Range of Correlated Transparent Conductors." *Laser & Photonics Reviews* **15**(5), 2000444 (2021). doi:10.1002/lpor.202000444
18. Y.-I. Kim¶, M. Jeong¶, **J. Byun**¶, S.-H. Yang, W. Choi, W.-S. Jang, J. Jang, K. Lee, Y. Kim, J. Lee, E. Lee, Y.-M. Kim*. "Atomic-scale identification of invisible cation vacancies at an oxide homointerface." *Materials Today Physics* **16**, 100302 (2021). doi:10.1016/j.mtphys.2020.100302
19. J. Jo¶, J. Byun, J. Lee, D. Choe, I. Oh, J. Park, M.-J. Jin, J. Lee*, J.-W. Yoo*. "Emergence of Multispinterface and Antiferromagnetic Molecular Exchange Bias via Molecular Stacking on a Ferromagnetic Film." *Advanced Functional Materials* **30**(11), 1908499 (2020). doi:10.1002/adfm.201908499
20. J. Jo¶, J. Byun, I. Oh, J. Park, M.-J. Jin, B.-C. Min, J. Lee*, J.-W. Yoo*. "Molecular tunability of magnetic exchange bias and asymmetrical magnetotransport in metalloporphyrin/Co hybrid bilayers." *ACS Nano* **13**(1), 894–903 (2019). doi:10.1021/acsnano.8b08689
21. S.-J. Kwon, B.-M. Jung, T. Kim, J. Byun, J. Lee, S.-B. Lee, U.-H. Choi*. "Influence of Al_2O_3 Nanowires on Ion Transport in Nanocomposite Solid Polymer Electrolytes." *Macromolecules* **51**(24), 10194–10201 (2018). doi:10.1021/acs.macromol.8b01603
22. T. T. Ly, G. Duvjir, T. Min, J. Byun, T. Kim, M. M. Saad, N. T. M. Hai, S. Cho, J. Lee*, J. Kim*. "Atomistic study of the alloying behavior of crystalline $SnSe_{1-x}S_x$." *Physical Chemistry Chemical Physics* **19**(32), 21648–21654 (2017). doi:10.1039/c7cp03481d

Preprint

- **J. Byun** (first author), K. Lee, M. Jeong, E. Lee, J. Bang, H. Kim, G. H. Gu*, S. H. Oh*. "Imaging 3D polarization dynamics via deep learning 4D-STEM." *arXiv:2506.06598* (2025). doi:10.48550/arXiv.2506.06598

CONFERENCE PRESENTATIONS

Talks

1. J. Byun, Z. Wang, S. Oh, J. Lee. (2021). "Novel Stabilization Mechanism on Polar Oxide Surface." 2021 Materials Research Society Fall Meeting.
2. J. Byun, Z. Wang, S. Oh, J. Lee. (2021). "Novel Stabilization Mechanism on Polar Oxide Surface." 2021 Korean Physics Society Fall Meeting.
3. J. Byun, J. Lee. (2021). "Giant Nonlinear Tunneling Current in HfO_2-based Anti-Ferroelectric Tunnel Junction." Electronic Materials and Applications Conference 2021.
4. J. Byun, J. Lee, T. Choi, J. Lee. (2021). "Interface engineered TER optimization in HfO_2-based FTJ." 2021 Korean Physics Society Spring Meeting.
5. J. Byun, J. Lee, T. Choi, J. Lee. (2020). "DFT investigation on structural stability on the ferroelectric HfO_2 on the electrode." 2020 Korean Physics Society Fall Meeting.
6. J. Byun, T. Min, J. Lee. (2020). "Nonlinear current-voltage behavior in HfO_2-based ferroelectric domain-wall tunnel junction." 2020 Korean Ceramic Society Spring Meeting.
7. J. Byun, T. Min, J. Lee. (2020). "Giant Nonlinear Tunneling Current in HfO_2-based Anti-Ferroelectric Tunnel Junction." American Physical Society March Meeting 2020.
8. J. Byun, T. Min, J. Lee. (2019). "First-principles studies of the effects of oxygen vacancies on the HfO_2-based ferroelectric tunnel junction." 2019 Materials Research Society Fall Meeting.
9. J. Byun, T. Min, J. Lee. (2019). "First-principles studies of the HfO_2-based ferroelectric tunnel junction." 5th International Conference on Advanced Electromaterials.

10. J. Byun, T. Min, J. Lee. (2019). “Design and modeling of optimal HfO2-based ferroelectric tunneling junction.” 2019 Korean Physics Society Fall Meeting.
11. J. Byun, T. Min, J. Lee. (2019). “Design and modeling of optimal HfO2-based ferroelectric tunneling junction.” Asia-Pacific PFM 2019 Conference.
12. J. Byun, T. Min, J. Lee. (2019). “Complex bandstructure calculation for the ferroelectric tunneling junction current.” 2019 Korean Physics Society Spring Meeting.
13. J. Byun, T. Min, J. Lee. (2018). “First-principles studies of the effects of effective mass on the HfO2-based ferroelectric tunnel junction.” 2018 Korean Physics Society Fall Meeting.
14. J. Byun, T. Min, J. Lee. (2018). “Novel high coercivity Fe16-xAlxN2 alloy.” IcAUMS 2018.
15. J. Byun, T. Min, J. Lee. (2018). “High coercivity Fe16-xAlxN2 alloy design.” Korean Magnetics Society 2018 Spring Conference.
16. J. Byun, T. Min, J. Lee. (2018). “Novel high coercivity Fe16-xAlxN2 alloy.” American Physical Society March Meeting 2018.
17. J. Byun, T. Min, J. Lee. (2017). “First-principles modeling of electron tunneling through ferroelectric tunnel junction.” 2017 KPS Fall Meeting.
18. J. Byun, T. Min, J. Lee. (2017). “Design of Fe16-xAlxN2 with high-coercivity.” 2017 KPS Spring Meeting.

Posters

1. J. Byun, J. Jo, I. Oh, J. Park, M.-J. Jin, B.-C. Min, J.-W. Yoo, J. Lee. (2019). “First principles study of magnetic exchange interaction in Metalloporphyrin/Co interface.” 3rd Workshop on Functional Materials Science.
2. J. Byun, T. Min, J. Lee. (2019). “First-principles studies of the HfO2-based ferroelectric tunnel junction.” 15th KIAS Electronic Structure Calculation Workshop.
3. J. Byun, T. Min, J. Lee. (2019). “Ab initio study of the role of effective mass in ferroelectric tunneling junction.” 15th Dielectric Union Symposium.
4. J. Byun, T. Min, J. Lee. (2019). “First-principles studies of the effects of oxygen vacancy and effective mass on the HfO2-based ferroelectric tunnel junction.” 10th APCTP Workshop on Multiferroics.
5. J. Byun, T. Min, J. Lee. (2018). “First-principles studies of the effects of oxygen vacancies on the HfO2-based ferroelectric tunnel junction.” 12th Japan-Korea Conference on Ferroelectrics.

TECHNICAL SKILLS

DFT/MD codes: VASP, Quantum Espresso, CP2K, Wannier90, Phonopy, GULP · **ML:** PyTorch · **Programming:** Python, C++ (Arduino) · **Scientific computing:** Linux server management, SGE, Git

OTHER

- Military service (completed)
- 2021 — President of the (Physics Dept.) Student Council
- 2018 — KIAS CAC Summer School on Scientific Computing and Machine Learning
- 2017 — Python study manager for graduate students